

Lecture 5: Intertemporal Choice & Savings Behaviour

Mental Accounting, Debt Traps, and the Retirement Crisis

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Opening Puzzle — Is Money Fungible?

Poll — Would You Spend These Differently?

Scenario A: You won **\$5,000** in the lottery.

Scenario B: You received a **\$5,000** tax refund from the CRA.

Scenario C: Your employer gave you a **\$5,000** year-end bonus.

In each case: how much would you spend vs save? Write down a number.

Standard Economics Says: Same Answer Every Time

\$5,000 is \$5,000. Rational agents should save/spend the same fraction regardless of source. But research shows people *systematically* treat these differently. Lottery winnings: spend. Tax refund: half-spend, half-save. Bonus: mostly save. This is **mental accounting**.

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2. **Mental Accounting** — money in separate “buckets”
3. **Sunk Cost Fallacy** — past costs that should be ignored
4. **The Debt Aversion Paradox** — holding debt and savings simultaneously
5. **The Canadian Savings Crisis** — detailed data
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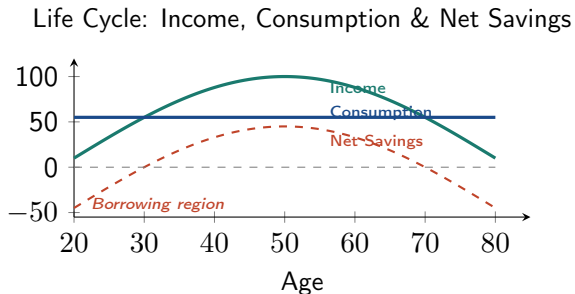
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The Life Cycle Hypothesis

Modigliani & Brumberg (1954): Rational agents maximise lifetime utility by **smoothing consumption**: borrow in **youth**, save in **prime years**, dissave in **retirement**.

Key Predictions

1. Temporary income shocks barely affect consumption.
2. MPC from windfall $\approx 1/T$ (spread over remaining life).
3. No response to predictable income changes.
4. Dissave in retirement.



All Four Fail

Consumption is **too sensitive** to current income, **drops sharply** at retirement, and saving rates are far too low vs LCH predictions.

Flavin (1981); Carroll (2001)

Consumption is too sensitive to current (predictable) income changes. If income is **fully predictable**, it should not change consumption (since it was already incorporated in the plan). But it does.

Canadian Evidence

- Tax refunds: Canadians increase spending by **25–50%** of refund in the refund month. Under LCH: should be $\approx 5\%$ (spread over 20-year horizon).
- Average Canadian tax refund (CRA, 2022): \$1,895.
- Retailers exploit this: “Tax Refund Events” in February–March. Canadian Tire, Walmart see **15–20%** sales spikes in refund season.
- Souleles (1999): US consumption rises sharply when Social Security cheques arrive — even though timing was 100% predictable.

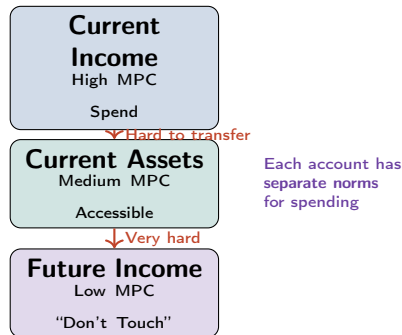
Mental Accounting — Thaler (1985, 1999)

People **categorise money into separate mental accounts** with different spending norms. **The Three Accounts (Shefrin & Thaler 1988):**

1. **Current Income:** High MPC — spend freely.
2. **Current Assets:** Medium MPC — savings, home equity.
3. **Future Income:** Near-zero MPC — RRSP. “Don’t touch.”

The Paradox

Households hold **credit card debt** at 19.99% *and* **RRSP** at 3%. Should pay off the card — but mental accounts prevent this.



The Sunk Cost Fallacy

Past costs that are **irretrievable** should not affect current decisions. Only future costs and benefits matter.

The Theatre Ticket

\$50 ticket; snowing; feel sick. Go?

If PAID: Most say yes (don't "waste" it).

If FREE: Most say no.

But \$50 is **gone either way**. Only relevant question: is going worth the discomfort?

Real Examples

- Bad relationship "after all the time invested."
- Holding a losing stock "because I can't sell at a loss."
- Finishing a terrible movie "because we paid."
- The Concorde: continued for sunk development costs.

[GAME] The Ski Trip Sunk Cost

Scenario A — Left Side of Room

One month ago, you bought a **non-refundable** ski lift ticket for \$80. Today, the day of the trip, you feel mildly sick and hear snow conditions are icy and dangerous.

Do you go skiing? Yes / No

Scenario B — Right Side of Room

One month ago, a friend gave you a **free** ski lift ticket (worth \$80). Same day: you feel mildly sick, conditions are icy.

Do you go skiing? Yes / No

Expected Results & Discussion

Far more students say YES in Scenario A. But the \$80 is gone either way. The only relevant question is: *Is skiing today worth the discomfort and risk?* If not in B, it should not be in A. The difference is the **sunk cost fallacy**.

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The Debt Aversion Paradox

Average Canadian household: **credit card debt** \$4,200 at 19.99% APR *and* **TFSA** \$18,000 at 3–4%.

The arbitrage: Pay off the card with TFSA savings = **guaranteed 17% risk-free return**.
Obvious — but mental accounts prevent it.

Gross, Souleles (2002)

US households: \$5K liquid assets + \$5K credit card debt at 18%. Should pay down. **Only 12%** do. “Emergency fund” and “credit card” are separate mental accounts.

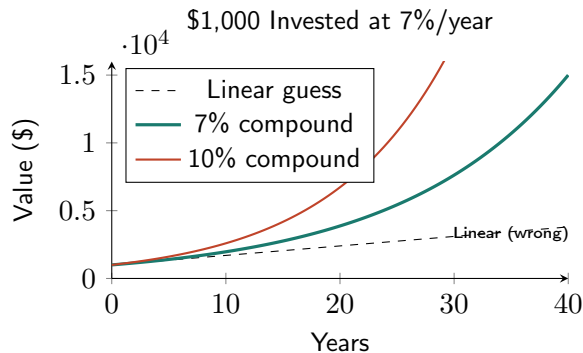
Behavioural Explanation

- TFSA = “emergency fund” (don’t touch)
- Credit card = “current spending”
- Transferring violates mental account norms
- Loss aversion: dipping in feels like a loss

Compound Interest Is Always Underestimated

Wagenaar & Sagaria (1975)

People systematically **underestimate compound growth**. They apply a *linear* mental model to what is an exponential process.



The Numbers

\$1,000 at 7% annual return:

- After 10 years: **\$1,967**
- After 20 years: **\$3,870**
- After 30 years: **\$7,612**
- After 40 years: **\$14,974**

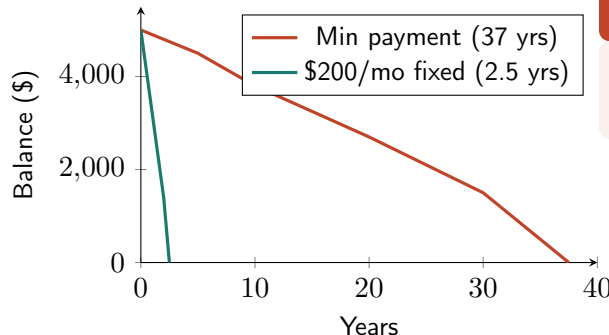
Student guess for 40 years: \$3,800.

Underestimate by **75%**. The “Rule of 72”:
money doubles every $72/r$ years.

The Minimum Payment Trap

\$5,000 at 19.99% APR. Minimum payment = 2% of balance.

Debt Repayment: \$5,000 at 19.99%



Min Payment: 37 yrs, \$10K+ interest

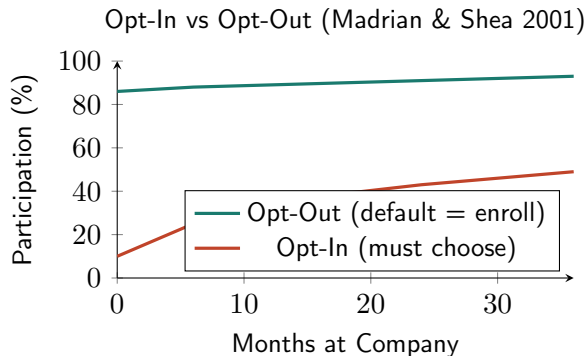
Pays off in **37 years**; total interest **\$10,233** — more than double the debt.

\$200/mo: Pays off in **2.5 years**; interest **\$1,065**; saves **\$9,168**.

FCAC (2021): Statements must now disclose payoff time and total interest $\Rightarrow$ higher payments up 12%.

Auto-Enrollment — The Power of Defaults

Madrian & Shea (2001): One US company changed 401(k) from **opt-in** to **opt-out**. Same plan, same rates — different default.



Same Plan. Different Default. Massive Difference.

- Opt-in: **49%** enrolled after 3 years.
- Opt-out: **86%** enrolled immediately; **93%** after 3 years.
- Mechanism: inertia + implicit endorsement.

Canada: Australia Superannuation = 11% mandatory (ultimate default). PRPP opt-in = low uptake. Federal pilot (2019): participation +40%.

Lusardi & Mitchell (2011) — The Big Three:

1. **Compound interest** (\$100@2%, 5 yrs: more than \$110?): 67% correct.
2. **Inflation** (1% savings, 2% inflation: buy more/less?): 75% correct.
3. **Diversification** (one stock safer than mutual fund?): 52% correct.

Canadian Data

- All three correct: only **42%** of Canadians.
- Correct answers predict **10× larger retirement savings**.
- **BUT** financial education programs: only +0.1% savings rate improvement (Fernandes et al., 2014). Information alone is insufficient.

Discussion Questions

- Q1. “Do you have a personal emergency fund? Is it in a separate account from your chequing account? Does mental accounting help you or hurt you here?”
- Q2. “Should Canada move to a mandatory defined-contribution pension system like Australia’s Superannuation (11% of every paycheck)? What are the trade-offs?”
- Q3. “Banks are required to display minimum-payment consequences on credit card statements. Is this enough? What else could be done?”

Key Takeaways — Lecture 5

- ✓ **LCH predicts consumption smoothing** — evidence shows excess sensitivity and undersaving.
- ✓ **Mental accounting:** money is NOT fungible across mental accounts; each account has different spending norms.
- ✓ **Sunk cost fallacy:** past costs affect current decisions even when they shouldn't.
- ✓ **Debt aversion paradox:** hold debt and savings simultaneously because they are in separate mental accounts.
- ✓ **Compound interest illusion:** people apply linear thinking to exponential processes — systematically underestimating savings growth.
- ✓ **Solutions:** auto-enrollment defaults, SMarT escalation, simplification, salience of long-run costs.
- ✓ **Canadian context:** RRSP gap, TFSA underutilization, credit card debt at record high, savings rate at historic low.

Next: Lecture 6: Social Preferences — We move from individual decisions to strategic interactions. Why do people give money away, reject unfair offers, and punish free-riders at personal cost?